turnbacktime

Biological Age Report

Prepared for: DOMINIC IFIDON

25th April 2022

Client ID: 9471

Security Key: DFB

Consultant ID: 41127

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1. Introduction

This report has been prepared for DOMINIC IFIDON on 25th April 2022

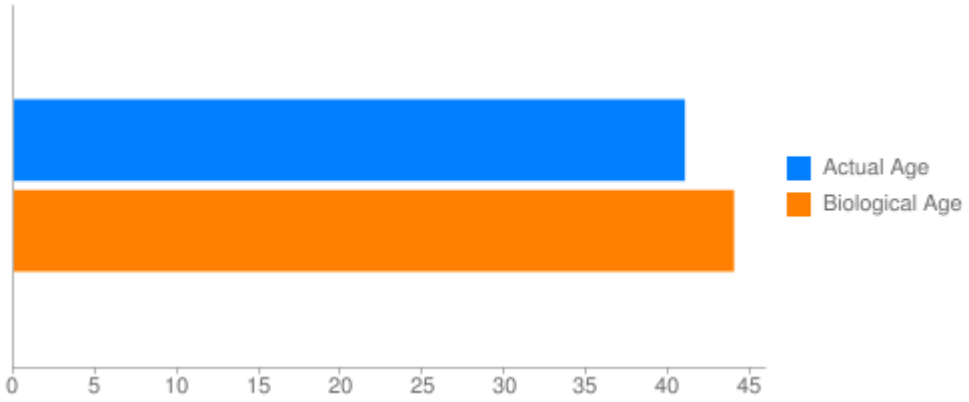
Date of birth:	21/04/1981
Age:	41
Gender:	Male
Height:	183cm
Weight:	100.00kg

The purpose of your report is not to provide you with a diagnosis of any medical condition. This report has been supplied to alert you to areas concerning your health that may need attention. It is important that you read the full report and consider the recommendations.

Regular testing as recommended by your Consultant is essential as health indicators can change rapidly and it is far better to be aware of these changes before it is too late to put them right.

2. Biological Age

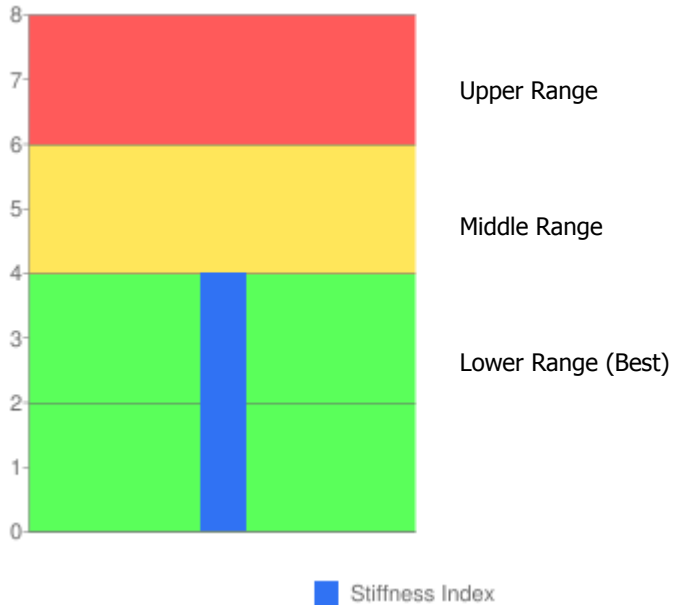
Your Biological Age Result: 44



Over 70% of people have a Biological Age that is higher than their chronological age, so your result is not unusual.You are advised to make an appointment with your Consultant for another test. This is because our health can change for the better or worse within surprisingly short periods of time. After a few tests you will start to see a trend on your Biological Age chart, and this will help you see if you are getting healthier or whether there are any areas that are starting to need some attention.

3. Arterial Stiffness

Your Arterial Stiffness Index reading for this test: 4



The average person will have a stiffness index reading of 4. Therefore you have had an acceptable result but we would ask you to remember that all of the results within this report are to be used as a guide and for reference only. The results are not intended as a diagnosis.

For a more information about Arterial Stiffness in general please see 'Appendix I - Arterial Stiffness Explained'

Stiff and narrow arteries restrict the flow of blood. 98% of the oxygen the body requires is carried by the haemoglobin within the blood. Therefore if the arteries are restricting the flow of blood, your skin, brain and other vital organs are not getting the amount of oxygen they need.

Although your result is acceptable, there is plenty of room for improvement. Restricting oxygen to vital organs can have a severe impact on health. Restricting oxygen to your skin can reduce elasticity and speed up the appearance of wrinkles, eye bags etc. Restricting oxygen to your brain can cause poor memory function and at extreme levels can contribute to strokes.

Restricting oxygen to your heart at extreme levels can cause a heart attack.

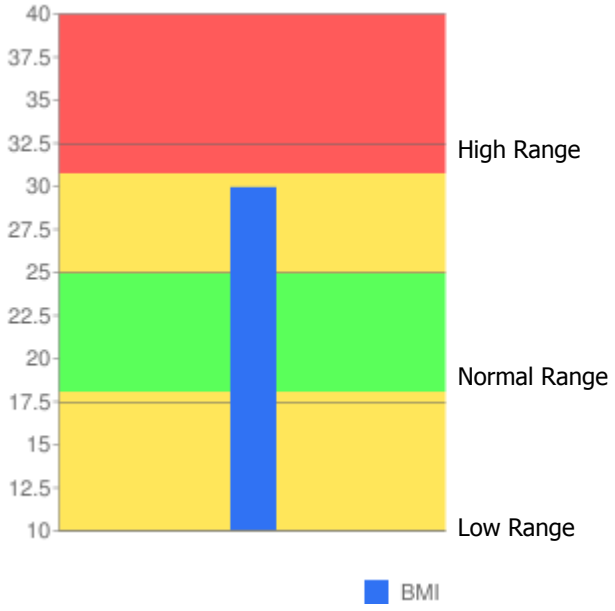
Restricting oxygen to your lungs can contribute to breathing difficulties.

4. Body Mass Index (BMI)

The Body Mass Index (BMI) is an accepted measure of body height to weight ratio. Human bodies range from around 15 (near starvation) to over 40 (possibly morbidly obese).

In the case of individuals who have a higher than average muscle mass, the BMI cannot be used as an accurate indicator of health. Therefore the charts should be used as a guide only.

Your BMI for this test: 29.9



Your BMI result indicates that you could be overweight. Please note however that BMI should only be used as a rough guide because it does not take bone or muscle mass into account.

If the high BMI reading is not due to high muscle mass or bone density, remember that even a minor weight loss (eg 10% of current weight) will help to reduce your risk of developing diseases associated with obesity.

Overweight patients who do not have a high waist measurement may need to prevent further weight gain rather than reduce weight.

4. Body Mass Index (cont.)

3.1 Recommendations

Talk to your doctor before starting any weight loss programme or supplement. Be wary of fad diets and rapid weight loss programmes. They may give you dramatic short term results but can be dangerous long-term. If you don't get enough calories, your body will go into starvation mode, your metabolism will slow down and weight loss will be even harder.

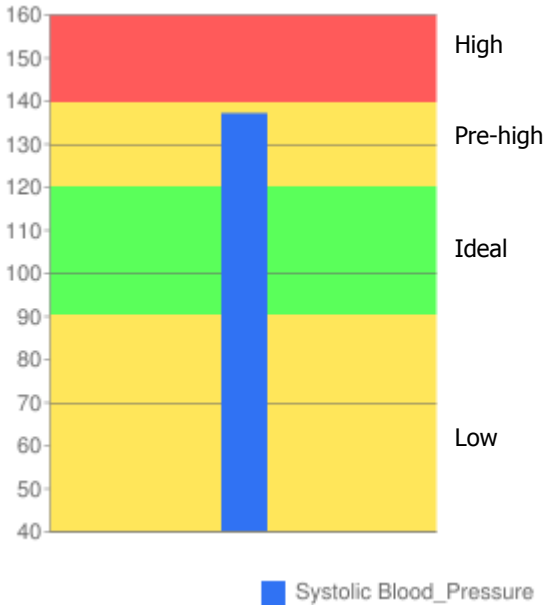
Be active. If you don't have much time for exercise, try to build activity into your daily routine, such as taking the stairs in place of a lift, parking further from your work building or the supermarket, or getting off the bus a couple of stops early. If you haven't been active, doing too much too soon can lead to injuries and burnout. Have a physical first and talk to your doctor about starting an exercise programme. You may want to start walking 10 minutes each day and gradually build up your time.

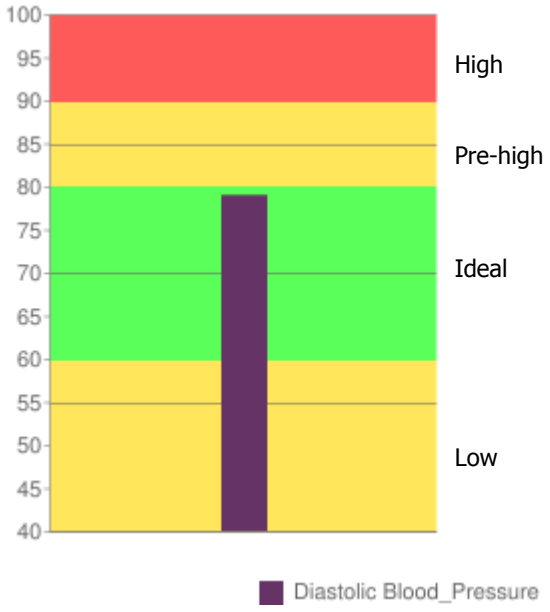
Set realistic weight loss goals, such as one or two pounds each week. Even a small amount of weight loss can make a huge difference to your health and losing weight gradually is more sustainable in the long term.

Know what you are eating. Try to limit your daily calories to less than 30% fat. Instead of junk food snacks, keep fruit, carrots or celery in your refrigerator.

5. Blood Pressure

Your Blood Pressure reading for this test: 137/79





Your reading indicates a normal blood pressure, but it is a little higher than it should be. You should try and lower it.

For more information about Blood Pressure in general please see 'Appendix II - Blood Pressure Explained'

5. Blood Pressure (cont.)

5.1 Recommendations

Here are some recommendations to help you maintain a healthy blood pressure.

1. Refrain From Eating Too Much Salt

Too much salt raises your blood pressure, so it is important to eat as little as possible. In fact, some people with high blood pressure may be able to avoid blood pressure medicines simply by cutting down on salt.

Most of the salt you eat is not what you add to your food, but is in prepared foods like bread, breakfast cereals and ready meals.

Don't add salt to food when cooking or at the table. When shopping for food, check the labels and choose low salt options when you can.

2. Eat more fruit and vegetables

Eating more fruit and vegetables helps to lower your blood pressure. Adults should eat at least five portions of fruit and vegetables every day. A portion is 80 grams, or roughly the size of your fist.

Try to eat a range of different fruits and vegetables. Dried, frozen and tinned are fine, but watch out for added salt, sugar or fats.

3. Keep to a healthy weight

Losing weight, if you need to, helps lower your blood pressure and reduce your risk of health problems. The best way to lose weight is to choose more low-fat and low-calorie foods, and increase your physical activity.

Set yourself realistic goals. Make small changes to your eating habits and activity levels that you can sustain.

4. Limit alcohol consumption

If you drink too much alcohol, this will raise your blood pressure over time. The current recommended limits are 21 units of alcohol a week for men and 14 units a week for women. A unit is roughly half a pint of regular strength beer or cider, a small (125ml) glass of wine, or a single pub measure (25ml) of spirits.

If you keep within the recommended alcohol limits, this should help keep your blood pressure down.

5. Stay Active

Being moderately active for 30 minutes five times a week can keep your heart healthy and lower your blood pressure. If you can't find 30 minutes in your day, increasing your activity by even a small amount can help.

Think about how you can be more active in your daily life. Any activity that leaves you feeling warm and slightly out of breath is ideal.

Appendix I - Arterial Stiffness Explained

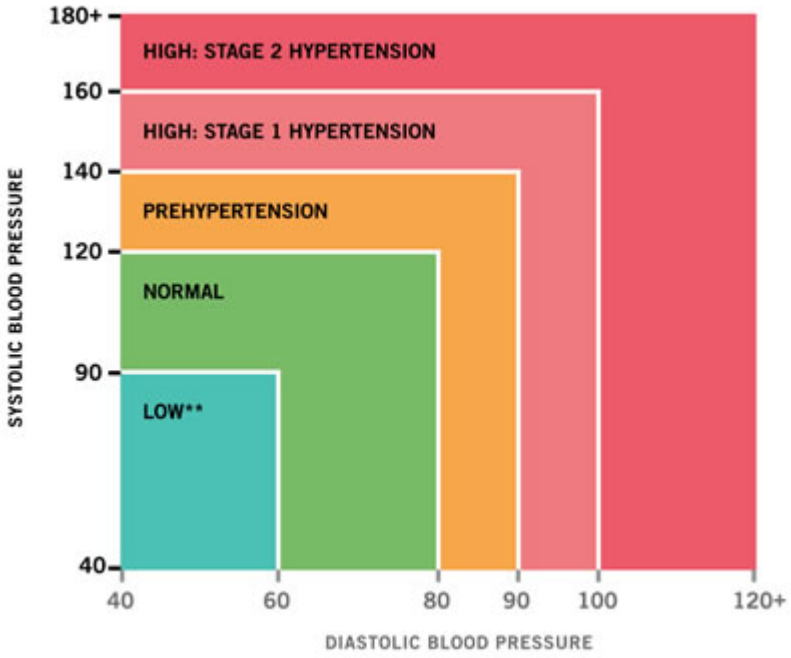
It has been appreciated for many years that there is a tendency for an individual's blood pressure to rise gradually with age. It was initially thought that this was simply due to a loss of elastic tissue from blood vessels resulting in a 'hardening' of the arteries. It was also recognised that this tendency in hardening was associated with an increased risk of strokes and heart attacks.

Our knowledge has advanced over the last 20 years and now we have a much better understanding of the mechanism underlying the rise in blood pressure. Blood vessels are not just simple tubes but complex organs. Through the release of a substance called nitric oxide, a thin lining on the inside of blood vessels (endothelium) controls how relaxed an artery is and therefore affects its elasticity. In times of stress, the artery can become very constricted and cause a rise in blood pressure. If the endothelium does not work properly, this can be associated with a condition called endothelium dysfunction that results in a loss of elasticity of the artery described above. This dysfunction has now been seen as a precursor to the development of ischemic heart disease (narrowing of the heart arteries with fat) and strokes. The Turn Back Time procedure is a test of this endothelial function and elasticity of the artery and expresses this as arterial stiffness.

Appendix II - Blood Pressure Explained

Blood pressure is the pressure of blood in your arteries. Your heart is your body's main pump and it pushes blood through your arteries by contracting and relaxing. When it contracts, it forces blood through your arteries and pushes the pressure of the blood to the highest point - the systolic pressure. When your heart relaxes between beats, your blood pressure is at its lowest - its diastolic pressure.

It should be noted that a minimum of two readings taken on separate occasions are required to establish your 'normal' levels.



To work out what your blood pressure readings mean, just find your top number (systolic) on the left side of the blood pressure chart and your bottom number (diastolic) on the bottom of the blood pressure chart. Where the two meet is your blood pressure.

- 90 over 60 (90/60) or less: You may have low blood pressure.
- More than 90 over 60 (90/60) and less than 120 over 80 (120/80): Your blood pressure reading is ideal and healthy. Follow a healthy lifestyle to keep it at this level.
- More than 120 over 80 and less than 140 over 90 (120/80-140/90): You have a normal blood pressure reading but it is a little higher than it should be, so you should try to lower it. Make healthy changes to your lifestyle.
- 140 over 90 (140/90) or higher (over a number of tests): You may have high blood pressure (hypertension). Change your lifestyle - see your doctor or nurse and take any medicines they may give you.

Appendix III - Lifestyle Notes

1. Hydration

Average glasses of water per day over the last 7 days: More than 5

You seem to be drinking a reasonable amount of water. Water is important for the body to function properly.

As a guide it is recommended to drink about 1.2 litres of water every day to avoid dehydration. This works out to be about six 200ml or eight 150ml glasses, cups or mugs. In hotter climates, the body will need more water than this.

2. Caffeinated Drinks

Average per day over the last 7 days: 3-5

Your caffeine intake could be reduced.

A study found that one cup of a caffeinated drink is enough to reduce blood flow to the heart by 22% within an hour of being drunk. The study was funded by The Italian Ministry of Education and the Associazione Onlus Nutrizione e salute, Italy. The paper was published in the peer-reviewed medical journal *The European Journal of Clinical Nutrition*.

3. Alcoholic Drinks

Average per day over the last 7 days: 3-6
(drink = 1 unit alcohol which is about 1 small (125ml) glass of wine or half pint of regular strength beer)

Your alcohol intake would seem to be a little higher than the current guidelines.

The recommended daily limits for alcohol consumption are:

- no more than three to four units a day for men
- no more than two to three units a day for women

For both men and women, it is also recommended to include some alcohol-free days each week. You are putting your health at risk if you regularly exceed the recommended daily limits. Too much alcohol is a significant risk factor in the development of heart disease.

4. Smoking

The fact that you are a non smoker is very good for your general health and especially the health of your heart and cardiovascular system.

You should be aware however that it is essential to avoid breathing cigarette smoke from others. When non-smokers breathe in second hand smoke - also known as passive smoking - it can be harmful. Research shows that exposure to second hand tobacco smoke is a cause of heart disease in non-smokers.

5. Medications

You mentioned to your Consultant that you are on medication. You should be aware that some medicines can cause higher than average arterial stiffness and blood pressure readings. However in your case, they don't seem to be having an effect in these areas. If you should notice a change, please speak to your doctor.

6. Pain

You told your Consultant that you were in some pain today. Temporary pain causes the body to produce chemicals that may temporarily increase your arterial stiffness. In your case, the fact you are in pain does not seem to have affected your arterial stiffness reading.

7. Stress

You indicated that you have been suffering from stress. We recommend that you return for further testing so that your arterial health can be monitored. Stress, and how we react to it, can directly affect our blood vessels. Even the normal stress that we experience in everyday life can cause a temporary deterioration in our vascular function, through hormonal changes and increases in the adrenaline in our bloodstream. These temporary changes can persist with chronic emotional stress, especially in the Type A individuals who dump adrenaline into their bloodstreams at the slightest provocation.. Chronic stress can also cause increased inflammation (sometimes producing elevated CRP levels), which is associated with an elevated risk of atherosclerosis.

8. Questionnaire Record

The following table summarizes your answers to the lifestyle questions for this test. This information might prove helpful in analysing your test results.

Time of Day	Between 14:00 and 17
Q1: Hydration - average glasses of water per day over last 7 days?	More than 5
Q2: Caffeinated drinks - average per day over last 7 days?	3-5
Q3: Alcoholic drinks - average per day over last 7 days?	3-6
Q4: Cigarettes smoked in past month?	No
Q5: Do you have any history of heart disease?	Yes
Q6: Are you on any medications?	Yes
Q7: Are you in any pain today?	Yes
Q8: Have you exercised or walked for longer than 30 minutes within the last 2 hours?	No
Q9: Would you describe yourself as suffering from stress over the last month?	Yes

Appendix IV - Nitric Oxide

Nitric Oxide is a gas molecule produced within the human body. Despite being a simple molecule, Nitric Oxide is an important biological regulator. It was proclaimed "Molecule of the Year" in 1992.

Nitric Oxide causes widening of blood vessels which is known as vasodilation. It results from relaxation of smooth muscle cells within the vessel walls, particularly in the large veins, large arteries, and smaller arterioles. The process is essentially the opposite of vasoconstriction, which is the narrowing of blood vessels.

When blood vessels dilate, the flow of blood is increased due to a decrease in vascular resistance. Therefore, dilation of arterial blood vessels plays an essential role in helping to maintain healthy arterial pressure and left ventricular filling pressure.

When Nitric Oxide becomes deficient, barriers break down and protection is lost, making the body more susceptible to high blood pressure, arteriosclerosis and heart disease. It's the body's natural way of maintaining arterial health - provided we make enough of it. It is important therefore to maintain healthy levels of Nitric Oxide at all times.

Supplements

With the modern diet, it is often necessary to increase the intake of natural foods that can assist the body produce healthy levels of Nitric Oxide. We have listed some useful supplemental foods below.

- L-Arginine
- Hawthorn Berry
- Allicin
- Beetroot
- Ginkgo Biloba
- Lycopene
- Omega 3

Further Testing

It is essential that a regular check be kept on your health. Each time you have a test, your results are added to our database. This will help you keep a progressive check on your Biological Age, arterial health, BMI and blood pressure.

To book a test for either yourself, a friend or family member, please contact your Consultant: Nnedimma Iwueke on +2349068187012

Please keep caffeine and alcohol to a minimum for at least two hours before your next test.